

AI decrypts manuscript

Discovered hundred years ago, the 240-page Voynich manuscript has been finally cracked by using artificial intelligence. <http://digit.in/AIvoynich>



Google closes deal on HTC

The \$1.1 billion deal has finally closed between the two companies with Google taking over most of HTC's smartphone design division. <http://digit.in/GoogHTC>

Keeping it COOL

Diving deep into the tech that keeps your gadgets 'oh-so-chilled'

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very generation of computing hardware increases the power of your devices to do amazing things

everyday – run your favourite games, bring powerful videography and post production capabilities to your pocket, or even take the brunt of your never ending array of open browser tabs. What also increases at the same time is the ease with which your device can heat up, thanks to more components like transistors being crammed into smaller areas, thus building up the heat they give off due to resistance. But then, shouldn't all kinds of devices become increasingly prone to overheating with every generation? The answer is no, thanks to the several types of cooling technology that are used across your devices and gadgets with an objective to cool them as efficiently as possible. Some rely on purely passive cooling, which uses natural convection and dissipation, while some opt for active cooling, which uses an external device to enhance cooling. But beyond this broad classification, there are numerous ways it's done across gadgets.

SMARTPHONES

Even the most rudimentary smartphones today boast of quite a few capabilities. And all those abilities come

at the cost of heating up the device when overused. There are quite a few advanced ways in which smartphone manufacturers are dealing with overheating, and some are yet to be applied to actual devices:

Heatpipes

Using pipes to dissipate heat in smartphones is not new. NEC did it way back in 2013 with the Medias X. The design involved a pipe that contained water (in miniscule quantities) that ran from the CPU to other areas of the smartphone. Once heated, the water drop would turn into water vapour, travel to other, cooler areas of the body and condense back into water, thus dispersing the heat before it travelled back to the CPU again. Some designs can skip water altogether and use dissipation to turn your entire smartphone into a heatsink. But it was Samsung that revolutionised the technology for its Galaxy S7 and S7 edge by designing a new pipe based cooling system that was 50 times more efficient than pure copper in terms of thermal conductivity. They were able to

achieve this by experimenting with alloys and metal mesh inside the pipe. Even in 2018, despite the availability of other types of tech Samsung has announced that they'll be using heat pipes in their 2018 models as well. The gaming centric Razor Phone also comes with a massive copper heatpipe



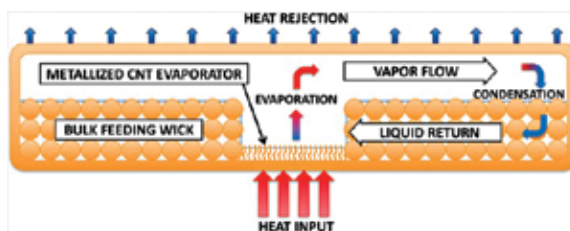
Source: iFixit

The heatpipe on the Galaxy S7

in its design. Right now, if one were to bet, heatpipes are likely to stay for another couple of years or at least until vapour chambers (explained later) become affordable.

Vapour Chambers

Working on the same principle of two phase cooling as heatpipes, vapour chambers are different from heat



How a vapour chamber works